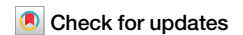


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Extensive glacitectonism and rapid deglacial fluid expulsion in the northeastern Barents Sea



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High-latitude regions store large amounts of carbon trapped by ice sheets, permafrost and gas hydrates, yet joint evolution of these climate components is poorly understood. Covered by a cold-based ice-sheet during the last glaciation, the seafloor of the northeastern Barents Sea reveals one of the largest known glacitectonic provinces in the Arctic. Up to 200-m high glacitectonic landforms are cratered and overlain by faults associated with ‘bright spots’ indicative of subsurface gas accumulations. However, sediment samples show low pore gas concentrations and no present-day gas seepage. We combine new observations and ice-sheet - hydrate modelling to propose that during the Late Weichselian, glacitectonism was caused by patchy substrate stiffening due to gradual growth of subglacial gas hydrates and permafrost. Ice decay led to rapid destabilisation and full drainage of shallow hydrate reservoirs and permafrost thaw, causing craterisation which was likely accompanied by large fluxes of carbon released into the water column. This study shows that these processes were more widespread across glaciated margins, also highlighting sensitivity and potential for abrupt changes of high-latitude carbon pools in response to complex interactions between the cryosphere, ocean, and solid earth.

Continental margins of the Arctic Ocean are estimated to contain as much as 20 Gt of organic carbon¹ that is trapped within permafrost-associated gas hydrates². The growth, decay, and geographic distribution of permafrost and gas hydrates are sensitive to changing environmental conditions, which affect the release and storage of large amounts of greenhouse gases in these deposits³. For instance, studies from continental margin offshore Svalbard have shown that even short-term, seasonal incursions of cold/warm water can modulate rates of gas hydrate dissociation and methane seepage activity⁴. On millennial timescales, changes to high-latitude hydrates and permafrost are driven by periodic build-up and disintegration of continental-scale ice sheets⁵. During past cold glacial periods, gas hydrates formed subglacially as pressure built up beneath growing ice sheets, while thick permafrost formed across vast shallow continental shelf areas that were emergent due to lower sea levels^{5,6}. Warmer interglacial periods were accompanied by decay of large ice sheets and rising sea levels, followed by rapid destabilisation of subglacial gas hydrates and thermal inundation of coastal permafrost^{7,8}.

Despite widespread high-latitude distribution of present-day hydrates and permafrost (Fig. 1a), geological indicators of their evolution have been observed only in areas that were either: (1) free of grounded ice during the

last full glacial period, such as the vast Canadian and Siberian Arctic shelf^{8–10}, or (2) formerly glaciated continental shelves where the ice sheet was warm-based, such as the western sector of the Barents Sea^{6,11,12}. In contrast, the shallow sector of eastern Barents Sea is believed to have been covered by a permanently cold-based ice sheet during the last glacial period, potentially allowing for simultaneous interactions between all three of these important climate components: ice sheets, hydrates, and permafrost¹¹.

Recent discovery of methane blow-out craters in the western Barents Sea² (Fig. 1a) suggests that abrupt deglacial fluid expulsion could have been triggered during deglaciation of hydrocarbon provinces, yet these processes have hitherto remained completely unknown in the entire vast eastern Barents Sea sector. In this study, we present, for the first time, high-resolution seafloor mapping from the shallow Barents Sea sector between Franz Josef Land and Novaya Zemlya, where relatively thin cover of Quaternary sediments directly overlies heavily faulted anticlines within eroded Cretaceous bedrock¹³ (Fig. 1b–d). We report the discovery of striking, heavily cratered glacitectonic features. By integrating geophysical, geological, and geochemical evidence with numerical thermodynamic modelling, we investigate the formation scenarios of the newly observed landforms in this poorly understood sector of the former Eurasian Ice Sheet (EIS).

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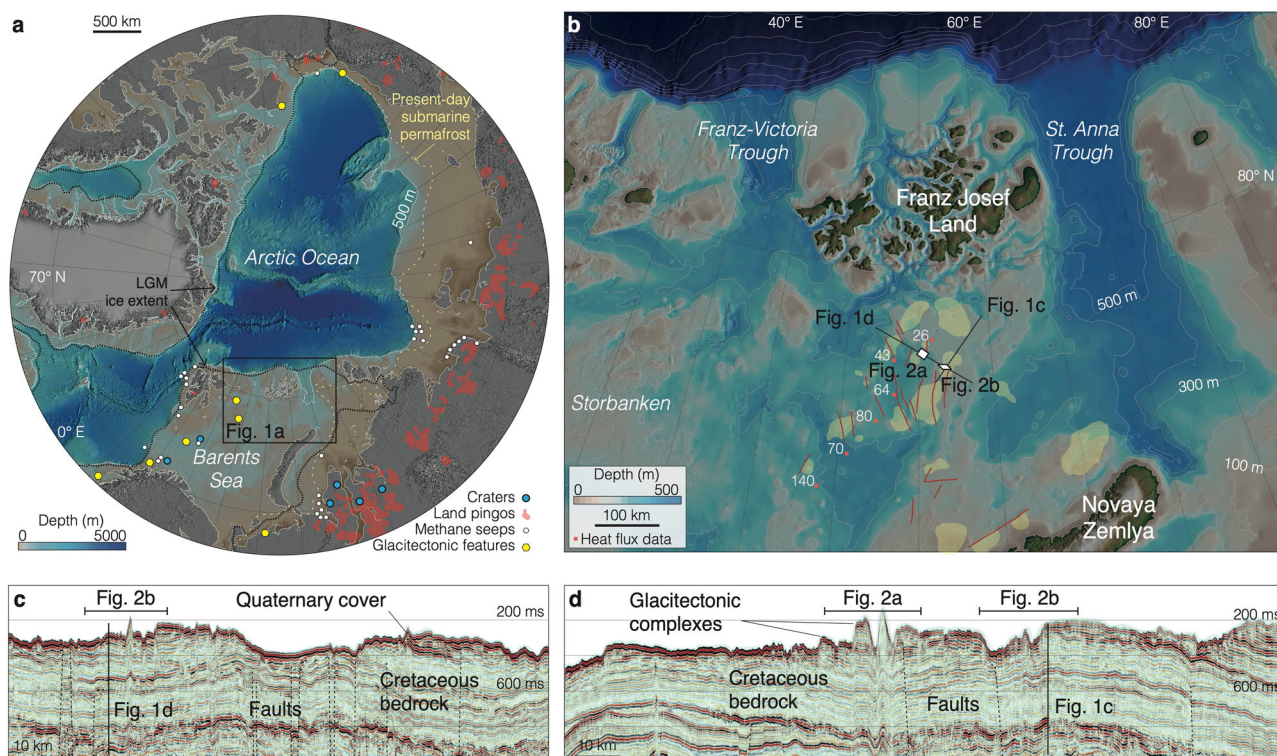


Fig. 1 | Arctic Ocean and the northeastern Barents Sea. a International Bathymetric Chart of the Arctic Ocean (IBCAO), v. 4.0⁶⁴. Black dashed line shows the Northern Hemispheric ice-sheet extent during the Last Glacial Maximum (LGM, ~25–21 ka)⁶⁵. Yellow dashed line shows the outer limit of present-day submarine permafrost¹. White dots show previously identified cold seeps releasing methane through the Arctic seafloor⁶⁶. Blue dots show giant blow-out craters in the Barents Sea² and on the Yamal and Gydan Peninsulas¹⁰. Yellow dots show the previous discoveries of glaciectonism in the Barents Sea¹⁷. Red areas indicate locations of

terrestrial pings as mapped in ref. 67. **b** Seafloor of northeastern Barents Sea showing the location of the study area. Annotated red dots show the past heat flux measurements (in mW m^{-2}) in the vicinity of the study area, available from the Global Heat Flow Database⁶². Potential hydrocarbon accumulation structures (traps) are shown by yellow semi-transparent areas¹³. Red segments mark some of the previously mapped faults in the study area¹³. **c, d** Two seismic lines showing the regional seismostratigraphic setting characterised by faulted bedrock across the study area.

Results

Geophysical observations

Multibeam imagery reveals an irregular, patchy seafloor morphology with several northwest-southeast elongated depressions that terminate with large sedimentary thrust-block hills at their southeastern ends (Fig. 2a). Depressions represent striking, 2–5 km wide, 4–8 km long, flat-bottomed overdeepenings that are incised up to 75 m into the seismically imaged bedrock (Fig. 2c). Thrust-block hills have a chaotic, semi-transparent internal acoustic character (Fig. 2b, c) and vary in configuration from rafted isometric hills to more complex, arcuate or lobate morphology. They are elevated 50–150 m above the surrounding seafloor; the elevation amplitude between individual depression and thrust-block hill can be as high as 200 m (Fig. 2a). The longest fully mapped depression is 0.31 km^3 in volume and shows slight widening at the terminus where it transitions into 0.28 km^3 hill that is slightly tilted relative to the depression's elongation axis (Fig. 2a, d). Based on their morphology and acoustic character, we interpret the newly discovered depressions and adjacent thrust-block hills as glaciectonic hill-hole pairs, previously identified in several formerly glaciated continental shelf locations^{14–17}. These features are produced by grounded ice through displacement of subglacial substrate and formation of depression (hole), and transportation of rafted material further down the flow, forming adjacent hill¹⁵.

Smaller-scale submarine landforms include ~100 m -wide, linear, and curvilinear depressions incised several metres into the underlying substrate (Fig. 2a, b). With characteristic berms on either side, these incisions are interpreted as iceberg ploughmarks, produced by keels of drifting icebergs beyond the margins of marine glaciers and ice sheets¹⁸. Additionally, over fifty large (mean diameter of 370 m) isometric

depressions with rims are incised as deep as 30 m into the surrounding sediment and bedrock. These craters cluster in water depths of 180–250 m and are found both inside and around glaciectonic 'holes'. Craters appear to cut across iceberg ploughmarks in a few locations where they intersect (e.g., Fig. 2b). Single beam echo-sounding performed onboard showed no distinctive high-amplitude flares emerging from the craters into the water above.

Deep seismic imaging reveals high-amplitude reflections, phase-reversed in comparison to the seafloor reflection, located 400–500 m below the seafloor. These phase-reversed 'bright spots' typically indicate subsurface gas accumulations¹⁷ and are intersected by a series of faults that rise through the bedrock stratigraphy and often reach the seafloor (Fig. 1c, d; 2c; 3b). Based on a grid of several seismic profiles intersecting the study area¹³, fault mapping shows alignment with the northwest-southeast direction of hill-hole pairs¹⁹ (Fig. 2a). Above the stratified bedrock, only seismically transparent facies are present (previously interpreted as Quaternary sediment cover²⁰). These facies are imaged in greater detail with a sub-bottom profiler (Fig. 3).

Shallow stratigraphy can be roughly characterised by two major acoustic units. The uppermost unit is a very thin, (<1 m thick), acoustically semi-transparent homogenous drape overlying a hummocky surface of the seismic unit below (Figs. 2d, 3a). The drape unit has a high-amplitude upper reflection and often is so thin that only this seafloor reflection is distinguishable on shallow acoustic profiles. The unit below varies considerably in thickness (1–10 m), has irregular, hummocky morphology and homogeneous high- to medium-amplitude acoustic structure (Fig. 3). This unit unconformably overlies sub-horizontal, flat-lying reflections that attenuate as overlying units thicken (Fig. 3).

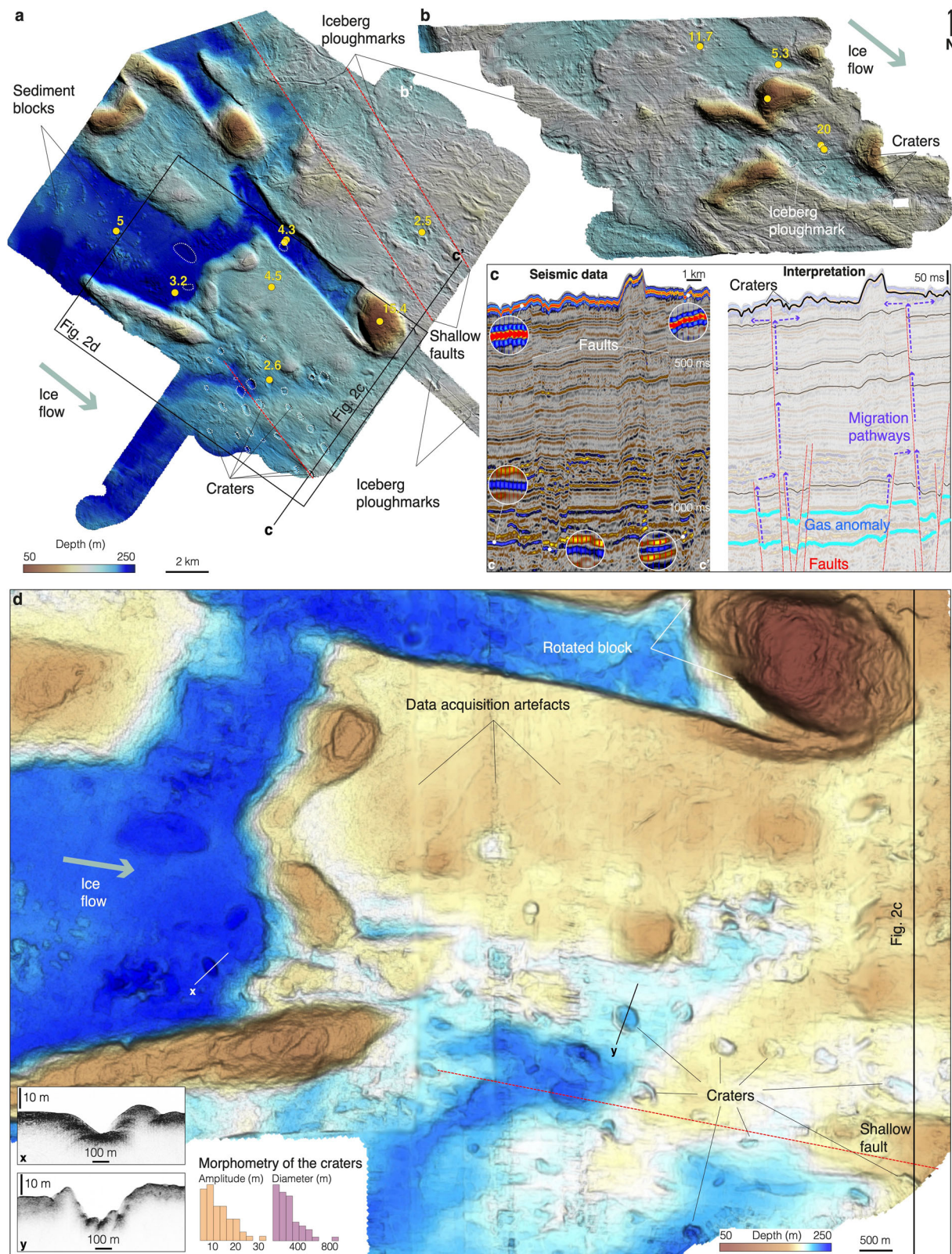


Fig. 2 | Glacitectonic landforms in the northeastern Barents Sea. a, b Bathymetric maps showing a series of rafted sediments and source depressions. Yellow numbers show the maximum value of methane concentrations (ppm) measured within the sampled sediment cores. **c** Seismic profile (on the left) and its interpretation (on the right) showing the structure, stratigraphy, possible fluid sources, and migration

routes underlying glacitectonic features and semi-circular depressions interpreted as craters. Reverse-phased 'bright spots' indicative of subsurface gas accumulations are shown in circular insets on the original seismic data (left panel). **d** Zoomed in sector of seafloor imagery presented on (a), showing dozens of craters. Insets on the bottom left show zoomed-in sub-bottom profiles crossing the craters.

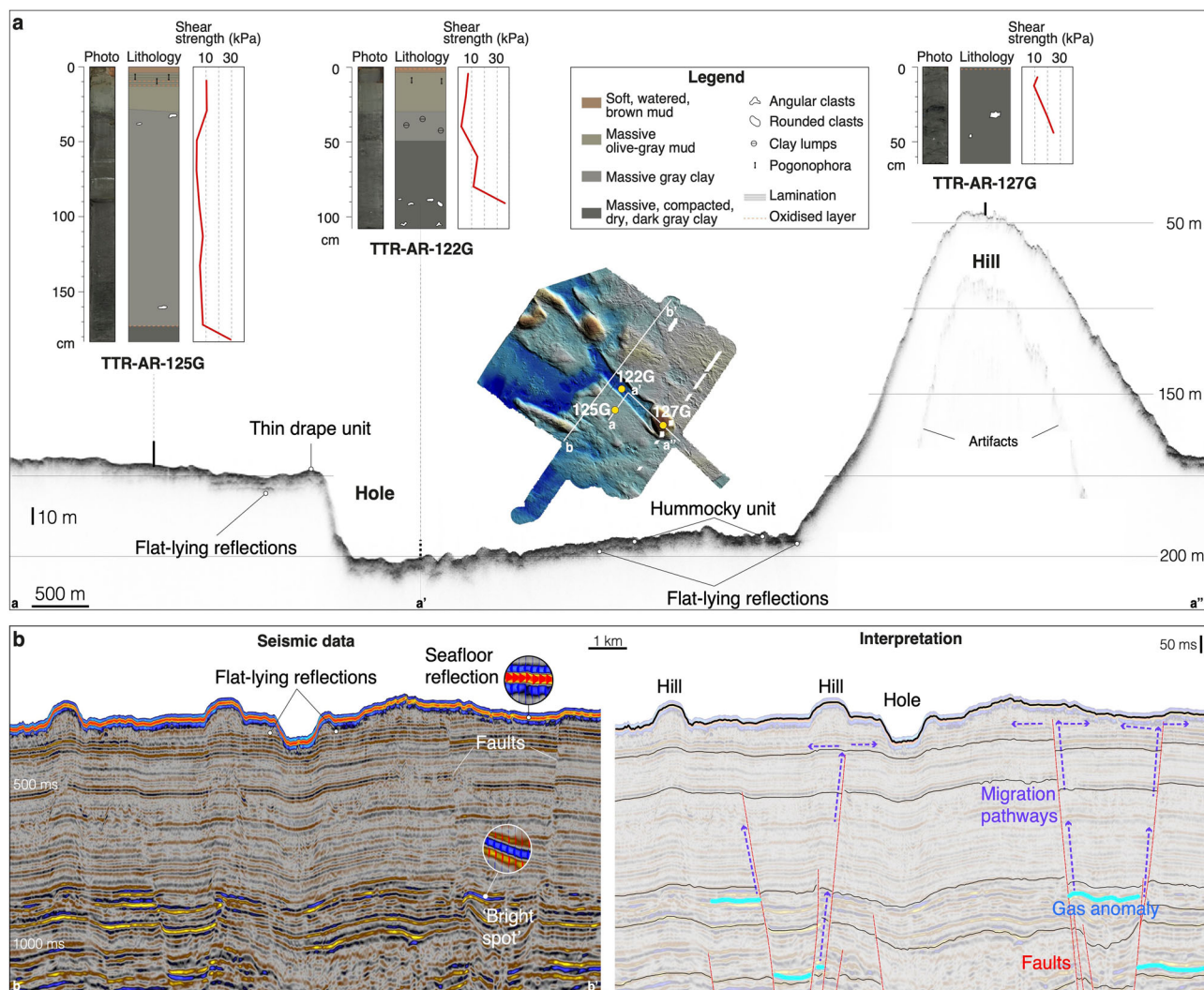


Fig. 3 | Shallow stratigraphy of the study area. a Simplified lithological logs and shear strength measurements of sediment cores collected along the major hill-hole pair in the northeastern Barents Sea. **b** Seismic profile (on the left) and its

interpretation (on the right) showing the structure and stratigraphy underlying overdeepened depressions and adjacent hills.

Core analysis

Thirteen sediment cores, up to 2 m long, were recovered from various locations across the study area (Fig. 2a). Cores were unable to penetrate further than 1 m at six coring locations, indicating hard bed below the uppermost sediment layers. With exception of three cores recovered from the tops of the major thrust-block hills, the uppermost sediment column generally consists of thin (several centimetres) watered layer of soft, brown oxidised mud, underlain by up to 70 cm-thick, olive-grey mud that is occasionally laminated and contains pogonophora tubes and interlayers of hydrotroilite (Fig. 3). These sediments are correlated with the uppermost drape unit on shallow acoustic profiles (Fig. 3) and are interpreted as a hemipelagic sedimentary cover accumulated upon full ice-sheet retreat, similar to open-marine Holocene sediments recovered on neighbouring Arctic shelf areas^{6,20,21}. Below the open-marine sedimentary drape are dry, compacted, massive, dark grey and grey clays with outsized angular to subrounded gravel and pebble-sized clasts of argillite and sandstone (Fig. 3). This lithofacies is correlated with the hummocky acoustic unit and interpreted as glacial and glacialmarine sediments.

Undrained shear strength is consistently very low (~3–5 kPa) in the uppermost hemipelagic layers. It tends to abruptly increase (up to ~40 kPa) with depth where the sediments change into massive compacted clays (Fig. 3). Geochemical examination of pore gas documents low methane concentrations in all recovered sediments (i.e., no more than 20 ppm, Fig. 2a, b).

Modelling results

Coupled with large-scale EIS model outputs from ref. 22 our hybrid ice sheet/gas hydrate simulations demonstrate a rapid increase in glaciostatic pressure and development of a subglacial GHSZ as soon as a grounded ice sheet glaciates the northeastern Barents Sea at ~28 ka (Fig. 4). The temperatures at the ice-bed interface stay below zero and decrease steadily, to -3°C and -9°C in the 'hot' and 'cold' heat flux scenarios, respectively. In such low-temperature conditions, the GHSZ attains at least 500 m thickness even in the 'hot' scenario. After ice retreats from the northeastern Barents Sea at ~13.5 ka, the lower GHSZ limit abruptly shifts ~300 m upwards, but its upper bound remains stable due to relatively high sea levels caused by a temporal lag in isostatic compensation (Fig. 4). As the glacio-isostatic rebound outpaces the eustatic sea-level rise, shallowing waters gradually reduce the hydrostatic pressure and cause a gradual thinning of the upper GHSZ limit in all heat flow scenarios (Fig. 4).

The timing and pace of GHSZ shrinkage after deglaciation also depend on the water column temperature at the seafloor. In the simulation shown on Fig. 4 sea bottom temperature was set to the in situ measurement value of -1.25°C . In the coldest scenario (sea bottom temperature of -1.25°C and heat flux value of 30 mW m^{-2}), the upper limit of the modelled GHSZ gradually drops starting from around 10 ka, reaching as much as 60 m below the present-day seafloor (as shown in Fig. 4). Increasing the bottom water temperature to 0°C results in an earlier

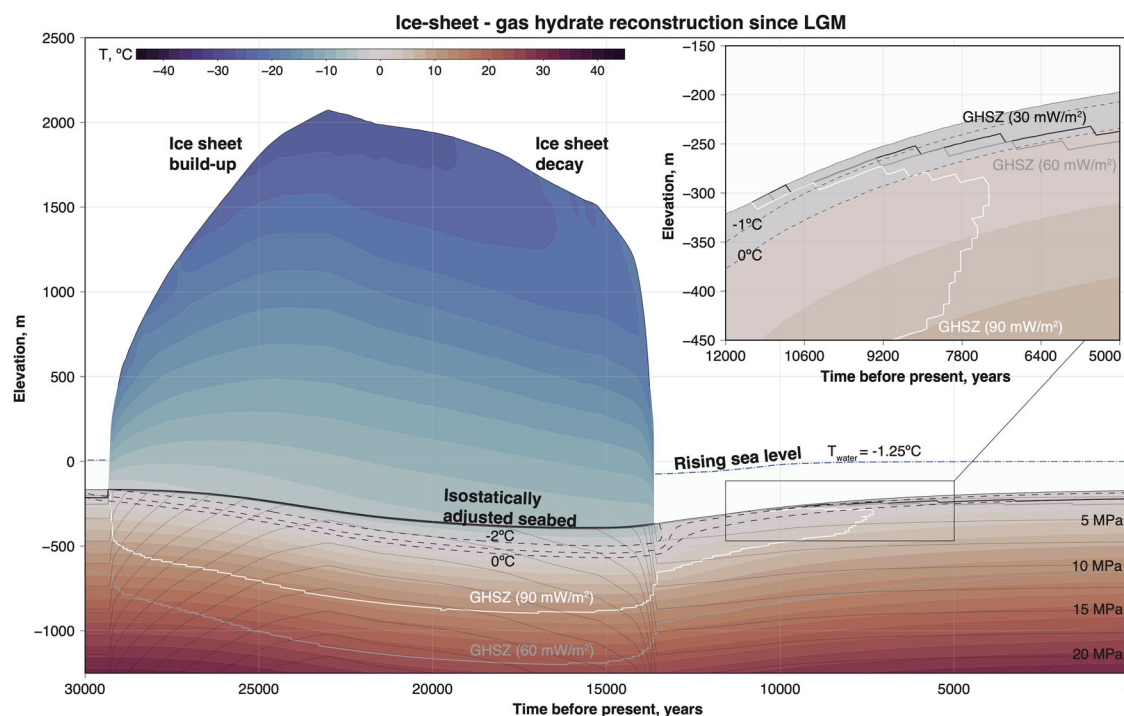


Fig. 4 | Simulated evolution of subsurface temperature, pressure, and GHSZ thickness throughout the last 30,000 years in the northeastern Barents Sea. The boundary conditions (e.g., ice thickness, seabed elevation, sea level) are from ref. 22.

Thin dashed black lines show isobars below the seafloor. Temperature distributions are shown for heat flux value of 60 mW m^{-2} .

(around 13 ka) and quicker shrinkage of the GHSZ, as well as a 50 m upward shift of the 0°C isotherm.

Discussion

Glacitectonism in the northeastern Barents Sea

Discovery of large glacitectonic features on the shallow seafloor 200 km south of Franz Josef Land has several glaciological implications that could help constrain existing ice-sheet models. Firstly, the ‘fresh’ topography and thin (less than 1 m) layer of overlying open-marine sediments indicate recent glacitectonism during the last glaciation. Extensive deformation of the seafloor in the study area may have taken place until as late as ~13 ka, when grounded ice fully retreated from the area¹¹. Secondly, their position on the shallow bank, not within a cross-shelf trough, and the lack of elongated, flow-parallel landforms that indicate a fast-flowing palaeo-ice stream suggest a slow ice-flow setting. This is supported by regional EIS models predicting low ice-sheet surface velocities in the shallow sectors of northeastern Barents Sea during the last deglaciation¹¹. Lastly, the northwest-southeast elongation of the holes and their position northwest of the hills indicates a southeastward direction of ice flow and corresponding glacitectonic deformation.

Different mechanisms for substrate deformation are proposed based on the varying morphology of glacitectonic complexes. Elongated holes that end with isometric hills of roughly equal width and volume suggest a focused, snowball-like formation process, possibly in an ice-marginal environment. In this scenario, a small fragment of sediment and underlying bedrock is rafted and thrust by the ice margin, growing in volume while retaining its width as it is pushed through the substrate along a single decollement. The fragment abruptly terminates several kilometres from its original detachment location, with no evidence of being overridden by grounded ice. In contrast, the irregular, arcuate or lobate hills with their smeared morphology and elongated parts that extend beyond the main sediment accumulation areas suggest subglacial deformation^{14,23}.

The seismically-imaged base of glacitectonic erosion is incised tens of metres deep into the bedrock, implying glacitectonic rafting, mixing and remoulding of subglacial sediment and shallow bedrock along a boundary

between high- and low-strength material. Such localised patches of substrate with high basal strength are also known as ‘sticky spots’²⁴. Theoretical models and observations from both contemporary and palaeo-ice-sheet beds suggest the following causes of sticky spot formation: (1) large bedrock bumps; (2) till-free areas; and (3) consolidated subglacial sediments^{24,25}. The largely smooth, sub-horizontal bedrock strata overlain by high-strength glacial sediment of varying thickness in the study area suggest that the latter (sediment consolidation) was a more likely cause of widespread basal ‘stickiness’. Although ‘sticky spots’ have been mostly related to fast-flowing ice streams, our results indicate their possible existence in areas occupied by relatively slow-flowing sectors of ice sheets.

Generally, porous subglacial substrate is consolidated when pore-water pressure is reduced²⁶. Three dewatering mechanisms have been previously described, first of them representing ‘water piracy’, whereby channelisation of initially distributed subglacial meltwater system leads to pore water removal from the surrounding sediment^{27–29}. Second, basal freeze-on causes extraction of pore water, ice cementing, and consolidation of subglacial substrate^{30,31}. Finally, stiffening of sediment and underlying bedrock can also occur due to formation of subglacial gas hydrates^{17,32}. No morphological evidence for channelised meltwater drainage in the study area (Fig. 2a, b) and the low-temperature, high-pressure conditions predicted by large-scale ice-sheet models¹¹ and our targeted thermodynamic simulations (Fig. 4) rule out the first dewatering mechanism. Therefore, basal freezing and/or the formation of subglacial gas hydrates were more likely causes of sticky spot formation in the northeastern Barents Sea^{17,25,33}.

Depending on pore water salinity, sediments freeze between -2°C and 0°C ³⁴. The -2°C isotherm is present subglacially in all modelled scenarios (Fig. 4) slowly propagating down to reach depth of 40 to 250 m below the ice-sheet base, depending on the geothermal heat flow. Large-scale ice-sheet models corroborate our simulation results, suggesting subglacial temperatures in the study area as low as -12°C during the LGM (e.g., Fig. S3 in ref. 2). Thus, long-term sub-freezing temperatures sustained over approximately 15,000 years of glaciation were suitable for the gradual growth of a permafrost layer underneath the grounded ice sheet in the northeastern Barents Sea (Fig. 4).

Freezing-induced pore water withdrawal could result in a four-fold increase in the shear strength, stiffening the subglacial sediments and the underlying bedrock³¹. Permafrost-related glacetectonism is consistent with findings from field investigations of glacier-permafrost interactions, which have shown that subglacial deformation can continue at sub-freezing temperatures through both proglacial thrusting and subglacial shearing^{23,35}. In addition, present-day submarine permafrost, up to 300 m thick, is widely present to the east of the study area in the southern Kara Sea. There, shallow shelf has been divided into areas of continuous and sporadic, patchy permafrost³⁶.

Beneath the ~2-km-thick ice sheet, the GHSZ would be more widespread than the permafrost layer, even if geothermal temperature gradients in the study area are high (Fig. 4). Phase-reversed, high-amplitude reflections hundreds of metres below the seafloor (Figs. 2c, 3b) are typical indicators of deep gas reservoirs^{17,37}. Fault-controlled and stratigraphic migration from these reservoirs into the GHSZ would lead gas to combine with pore water and form patches of stiffened hydrate-bearing sediments and bedrock¹⁷. Furthermore, it has been suggested that geological settings with higher permeability (e.g., due to the presence of faults) are more prone to hydrate growth³⁸.

The hydrate growth rate from dissolved phase methane can range from hundreds to tens of thousands of years for hydrate accumulations to reach saturations that are detectable by geophysical methods³⁸. Such a broad, two orders of magnitude range of hydrate formation timescales is explained by drastically different advection rates (i.e., from mm/year to cm/year) observed in various hydrate provinces around the world^{39,40}. Although our data do not provide an estimation of possible past advection rates in the northeastern Barents Sea, the timescales of previously observed hydrate growth are generally compatible with the ~15 k.y. long period during which the study area was covered by a thick, grounded ice sheet.

Hydrates begin forming at the gas-water interface and eventually create networks or frame structures throughout the substrate, enhancing the shear strength and making them significantly more ductile compared to the brittle-like behaviour of deformation within pure permafrost³¹. Geo-mechanical experiments have shown that unfrozen sediment with 22% methane hydrate is as strong as permafrost with 85% ice in the pore space. The coexistence of hydrate and permafrost results in a further ~50% increase in shear strength due to hydrate and ice grain cementation³¹. Rock physics experimental models of hydrate-bearing sands demonstrated that even a small (i.e., 2–3%) increase in pore hydrate saturation could lead to abrupt increase in acoustic compressional velocity and shear strength⁴¹. Sediments with 60% hydrate saturation recovered from the Mackenzie Delta exhibit a shear strength of 6.7 MPa, which is much higher than that of subglacial till recovered from present-day ice streams in Antarctica, where the basal shear strength is typically less than 20 kPa^{17,26,42–44}.

Origin of large seafloor craters

Seafloor morphology in the study area clearly shows that craters overprint, and therefore postdate glacetectonic landforms. Crater rims appear to incise into iceberg ploughmarks, implying that the formation of craters followed cessation of iceberg activity in the study area. Thus, we infer the craters to have formed after the region fully deglaciated ~13–12 ka⁴⁵.

What might have caused postglacial craterisation in the shallow shelf areas of the northeastern Barents Sea? Models of subsurface crater formation typically require generation of pore-fluid overpressure. For instance, formation of 40 m wide seafloor crater in the Norwegian North Sea was observed after water injected at high pressure into deep sandstone reservoir leaked to the seafloor, causing fracturing and rapid formation of seafloor depression in a matter of several weeks⁴⁶. Recent work from Bjørnøyrenna cross-shelf trough, located about 800 km southwest of the study area, found that over-pressurisation of shallow sedimentary bedrock and subsequent gas blowout could be caused by the dissociation of methane hydrates during ice sheet retreat². The same mechanism explains the formation of pingos on the Svalbard continental shelf and permafrost craters on the Yamal and Gydan peninsulas^{3,10}. Unlike long process of methane hydrates formation,

their dissociation takes place on much shorter timescales (i.e., days to years)³⁸, making it a plausible cause of rapid gas blowout.

The dimensions, morphology, and clustered distribution of the circular depressions discovered in the northeastern Barents Sea closely resemble methane blowout craters. In the western Barents Sea, these features were reported from Bjørnøyrenna, located at a water depth of ~350 m near the present-day limit of the GHSZ, where active gas seepage is taking place^{2,47}. In our study area, located well above the present-day GHSZ, the lack of gas flares in the present-day water column and the low levels of methane found in shallow sediments suggest that there is no ongoing focussed discharge of methane. The lack of active seepage could be explained by possible deactivation of faults due to isostatically-driven vertical crustal movements, as previously proposed offshore Svalbard⁴⁸. Furthermore, highly compacted tills overlain by fine open-marine Holocene mud cover together with shallow microbial oxidation likely act to significantly reduce possible seepage from subsurface⁴⁹. Together, these observations point towards a possible scenario where shallow hydrate reservoirs existed in the study area in the past but were fully drained through short-lived episodes of rapid deglacial gas expulsion that led to blow-out crater formation (Fig. 5).

Order-of-magnitude volumetric calculation suggests individual craters to host $0.1\text{--}5 \times 10^5 \text{ m}^3$ of hydrate, equivalent to $0.2\text{--}9 \times 10^8 \text{ m}^3$ of gas in normal conditions⁵⁰. This is only a fraction of global present-day cold seep flux of $2.5\text{--}3.5 \times 10^{10} \text{ m}^3/\text{y}$ ⁵¹; however, catastrophic bubble release from shallow seafloor, unlike slow seepage where methane dissolves in the water column, could deliver methane into the atmosphere^{52,53}. Timing of such methane release would vary for craters located at different depths, and depends on the deglacial oceanographic evolution of the eastern Barents Sea³.

An alternative cryovolcanic model of land crater formation on Yamal Peninsula suggests pore-pressure build-up due to refreezing and shrinkage of sealed intra-permafrost taliks - layers or bodies of unfrozen ground surrounded by permafrost^{9,54}. In submarine settings, groundwater that thaws permafrost at depth but refreezes as it ascends towards the seafloor can result in formation and subsequent collapse of open-system ice-cored pingos, as demonstrated by repetitive high-resolution multibeam observations in the Canadian Beaufort Sea, where present-day bottom waters are 1.4 °C below zero⁵⁵. Open-system pingos may also form as a result of pressurisation caused by uplift-induced permafrost expansion in former shallow marine sediments. Observed in Svalbard fjords, such pingos provide pathways for the focused release of methane-charged brackish waters^{56,57}.

Sub-zero bottom water temperatures measured in the study area suggest that permafrost-related mechanisms of crater formation also cannot be completely ruled out (Fig. 5). Water of 20‰ salinity is at the freezing point at -1.1 °C ^{34,55}. Core measurements in the Canadian Beaufort Sea showed considerable freshening of shallow groundwater - up to 15‰ within just a few metres below seafloor. A similar drop in salinity in the study area would, in principle, enable groundwater refreezing close to the seafloor, followed by heaving and pingo collapse as described by⁵⁵. However, the absence of a large thermokarst region like the one in the Beaufort Sea suggests that this mechanism in our study area would require spatially localised groundwater flow conduits and/or heat flux gradients that vary laterally.

The release of groundwater within submarine permafrost can lead to talik formation, as observed in the East Siberian Arctic, but likely requires complex pore water salinity distributions⁵⁸. The current rate of submarine permafrost degradation, 14 cm/year in the East Siberian Sea⁵⁸ is equivalent to hundreds of metres over millennia, providing sufficient time for a talik to form during deglacial submarine permafrost inundation. However, cryovolcanism in the northeastern Barents Sea would need expanding permafrost after talik formation, implying cooling of ocean bottom water. Past temperature reconstructions are absent in the northeastern Barents Sea, but investigations of the Svalbard continental margin have demonstrated that ocean waters experienced pronounced temperature fluctuations during the Holocene, with several incursions of warm Atlantic Water and subsequent cooling^{59,60}. Thus, past ocean temperatures remain an important but poorly

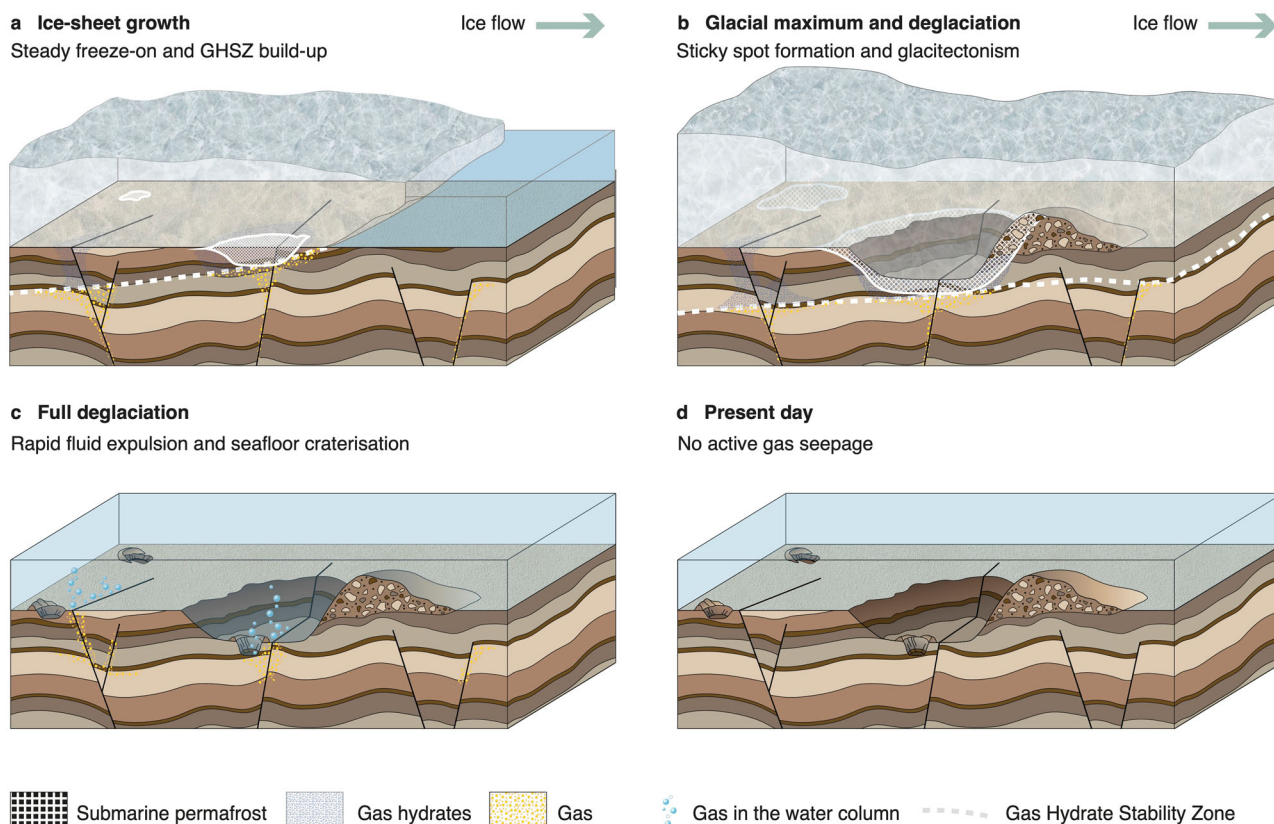


Fig. 5 | Hypothesised model of glaciectonism related to evolution of subglacial gas hydrates and permafrost on the shallow northeastern Barents Sea over the last glacial-interglacial cycle. a, b Build-up of cold-based ice-sheet enables the growth of subglacial gas hydrates and permafrost, causing extensive glaciectonism due to patchy stiffening of the subglacial substrate. **c** Following deglaciation, rapid

destabilisation of hydrate reservoirs and thawing of permafrost layer led to extensive seafloor craterisation and abrupt fluid release into the shallow water column. **d** Full reservoir drainage results in very low present-day pore gas concentrations and no active seepage through the water column.

understood variable that would improve our understanding of the subsurface environmental changes in the northeastern Barents Sea.

Conclusions

The discovery of a heavily cratered glaciectonic region on the shallow continental shelf of the northeastern Barents Sea suggests complex interactions between the cryosphere, ocean, and solid earth during the last glacial-interglacial cycle. Synthesising a hybrid ice sheet/gas hydrate model with geophysical, geological and geochemical observations, we test plausible conditions that caused glaciectonism and crater formation. We propose a scenario that is consistent with our data and modelling, where the growth of gas hydrates and permafrost beneath the ice sheet led to patchy stiffening of the subglacial substrate and glaciectonism. Following deglaciation, rapid dissociation and full drainage of hydrate reservoirs and thawing of permafrost layer led to extensive seafloor craterisation and abrupt fluid release into the shallow water column. These results highlight rapid evolution of subglacial carbon pools that are located beneath cold-based ice sheets, particularly at their shallow beds.

Methods

Geophysical, geological, and geochemical data

During the 2020 Training-through-Research (TTR-19) expedition to the northeastern Barents Sea, two blocks of multibeam swath-bathymetry data (Fig. 2a, b) were acquired within a ~2500 km² shallow shelf sector, the seafloor of which is characterised by highly irregular morphology¹⁸. The multibeam dataset covering a total area of ~150 km² was collected on board the R/V Akademik Nikolaj Strakhov at water depths ranging from 50 to 250 m. A Reson SeaBat 7150 multibeam echo-sounder was utilised for high-resolution seafloor mapping, with a configuration of 12 kHz and 256 beams

in a 1.5° × 1.5° grid. The MiniSVP profiler was used to calibrate the sound velocity and gather temperature and pressure data in the water column. The multibeam data were processed using PDS2000 software and resulted in images with a 15 m grid cell size. Sub-bottom shallow acoustic data were collected using a hull-mounted EdgeTech 3300 4.2 kHz profiler, along several lines across the multibeam survey areas. In addition, regional multichannel seismic profiles (Figs. 1, 2) acquired by JSC MAGE on the eastern flank of the Northern Barents Basin were used^{13,19}. These data were acquired in 2005–2007 using a 6-km-long streamer and a group of 37 sleeve airguns with cumulative volume of 3410 in ref. 3. The source frequency was within the range of 20–100 Hz.

Gravity cores were recovered and described from several sites across the study area (Fig. 2). Shear-strength measurements were performed every 20 cm using a torvane. Sediment samples collected every 20 cm were also examined for evidence of pore gas at each site. Approximately 40 mL of sampled sediment were placed into 120 mL vial containing 40 mL of solution of NaCl. Sealed vials were then injected with helium and shaken for at least 1 h at 350 rpm. The extracted gas samples in 20 mL vials were delivered to the onshore laboratory, where they were analysed using gas chromatograph “Chromatec Gasochrom 2000”.

Temperature and gas hydrate stability modelling

To examine the evolution of pressure and temperature through bedrock and ice column, we developed a finite-difference thermal model that solves the vertical diffusion-advection equation (Eq. (1)) at each one-year time step⁶¹:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial z^2} - w \frac{\partial T}{\partial z} \quad (1)$$

where t is time, z is height above the ice-sheet bed, w is vertical ice velocity (positive upwards) and α is thermal diffusivity:

$$\alpha = \frac{k}{\rho c} \quad (2)$$

where ρ is density, k is the thermal conductivity and c is the specific heat capacity.

A one-dimensional vertical spatial domain extending from the bedrock base to the ice surface was discretised with a minimum of 50 cells within the ice column, and 200 bedrock cells with constant 10 m vertical spacing. Surface air temperature, ice-sheet thickness, as well as sea-level and bedrock elevation history were sourced from the large-scale EIS model output, extracted specifically for our study area location and adjusted to its average present-day water depth of 176 m^{11,22}. Heat flux data were provided by the global heat flow database⁶².

Multiple simulations were run to account for the varying heat flux in the study area (Fig. 1b), using a range of possible heat flow values (i.e., ‘hot’, ‘warm’, and ‘cold’ scenarios, in which the heat flux is prescribed to 90, 60, and 30 mW m⁻², respectively). Computed subsurface pressure and temperature at each time step and every grid point were then compared against the theoretical hydrate stability phase diagrams generated using the well-established CSMHYD program⁶³ assuming pure methane and water salinity of 35‰⁵.

Data availability

The geophysical data presented in this manuscript is not publicly available due to commercial proprietary reasons and related data sharing permission restrictions for all primary data collected in the exclusive economic zone of Russian Federation, as imposed by Ministry of Science and Higher Education (contract π H-20-09/4H from 19th May 2020). To request access to the data, contact Dr Grigori Akhmanov: akhmanov@geol.msu.ru.

Code availability

The code related to this article is available online at: <https://github.com/sashamontelli>.

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Author contributions

Conceptualisation: A.M. Methodology: A.M., M.S., G.A. Investigation: A.M., M.S., G.A. Visualisation: A.M., M.S. Writing and editing the manuscript: A.M., M.S., G.A.

Competing interests

The authors declare no competing interests.

Additional information

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